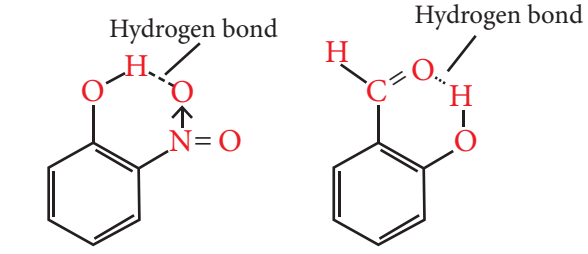
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structure of proteins and DNA. When a hydrogen atom (H) is covalently bonded to a highly electronegative atom such as fluorine (F) or oxygen (O) or nitrogen (N), the bond is polarized. Due to this effect, the polarized hydrogen atom is able to form a weak electrostatic interaction with another electronegative atom present in the vicinity. This interaction is called as a hydrogen bond (20-50 $kJ\ mol^{-1}$) and is denoted by dotted lines (…).

It is weaker than covalent bond (> 100 $kJ\ mol^{-1}$) but stronger than the van der Waals interaction (< 20 $kJ\ mol^{-1}$). Hydrogen bond has profound effect on various physical properties including vapour pressure (H_2O and H_2S), boiling point, miscibility of liquids (H_2O and C_2H_5OH), surface tension, densities, viscosity, heat of vaporization and fusion, etc. Hydrogen bonds can occur within a molecule (*intramolecular hydrogen bonding*) and between two molecules of the same type or different type (*intermolecular hydrogen bonding*).

Intramolecular Hydrogen Bond

Intramolecular hydrogen bonds are those which occur within a single molecule.



Ortho-Nitrophenol Salicilaldehyde

Figure 4.6 Intramolecular hydrogen bonding

Intermolecular hydrogen bond

Intermolecular hydrogen bonds occur between two separate molecules. They can occur between any numbers of like or unlike molecules as long as hydrogen donors and acceptors are present in positions which enable the hydrogen bonding interactions. For example, intermolecular hydrogen bonds can occur between ammonia molecule themselves or between water molecules themselves or between ammonia and water.

Water molecules form strong hydrogen bonds with one another. For example, each water molecule is linked to four others through hydrogen bonds. The shorter distances (100 pm) correspond to covalent bonds (solid lines), and the longer distances (180 pm) correspond to hydrogen bonds (dotted lines).

In ice, each atom is surrounded tetrahedrally by four water molecules through hydrogen bonds. That is, the presence of two hydrogen atoms and two lone pairs of electron on oxygen atoms in each water molecule allows formation of a three-dimensional structure. This arrangement creates an open structure, which accounts for the lower density of ice compared with water at 0°C. While in liquid water, unlike ice where hydrogen bonding occurs over a long-range, the strong hydrogen bonding prevails only in a short range and therefore the denser packing.

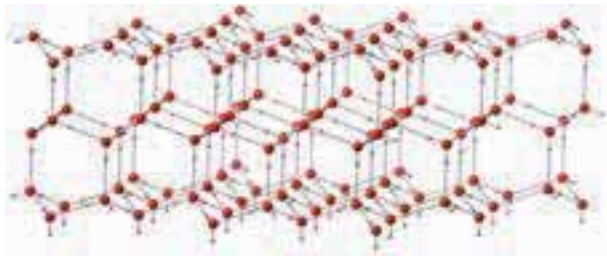


Figure 4.7 (a) Structure of Ice